Paraj Ganchaudhuri

Ph.D. Candidate (PMRF Fellow)
Systems, Control, and Automation
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About

Paraj is a Prime Minister's Research Fellow (PMRF) at the Indian Institute of Technology Guwahati, working on building algorithms for energy-efficient quadrotor flights by integrating optimal trajectory generation with advanced controller tuning. His research integrates optimization, optimal control, and model predictive control, among other approaches, to enhance quadrotor endurance. With strong interests in robotics, control systems, and emerging technologies, Paraj is passionate about building solutions from the ground up and translating research into impactful engineering applications. He thoroughly enjoys technical research, teaching, and maintains active interests in aviation and finance.

Education

Degree	Specialization	Institute/Board	Percentage	Year
Ph.D.	Systems, Control & Automation	Indian Institute of Technology Guwahati	92.5	2020 - Present
M.Tech	Control Systems Engineering	Birla Institute of Technology Mesra	88.3	2020
B.Tech	Electrical Engineering	National Institute of Technology Agartala	77.0	2017
H.S	Science	Indian School Certificate Examination	92.0	2013

Experience

- **Indian Institute of Technology Guwahati** *2020 - Present*
  Ph.D. Researcher and Teaching Assistant Guwahati, India
  – Research objective: Optimizing energy consumption in quadrotors for improving flight time.
  – Engaged in developing and teaching courses in Control Systems, Linear Algebra & Optimization, etc.
  – Involved in guiding and mentoring M.Tech, B.Tech students and research interns.

Projects

- **Ph.D. Research Work** *2021 - Present*
  Title: *PID and MPC tuning methods for quadrotors navigating energy-efficient trajectories* Github
  Supervisor: Dr. Chayan Bhawal
  – This work aims to find solutions towards energy efficient quadrotor missions by combining optimal control theory with controller tuning involving PID, Sliding Mode, and Model Predictive Control architectures.
  – Publications: 1 journal published in *Aerospace Science and Technology*, 1 to be submitted in *IEEE Transactions on Vehicular Technology*, 4 conferences.
- **M.Tech Research Work** *2019 - 2020*
  Title: *Controller design for a 2-Link manipulator with external disturbances and parametric variations* Github
  Supervisor: Dr. Sarbani Chakraborty
  – In this project, a robust Sliding Mode Control strategy has been derived for a 2-link robotic arm to ensure resilience against parametric uncertainties and external disturbances.

Technical Skills

- **Programming:** Python, Java, C, MATLAB, LaTeX.
- **Tools/Frameworks:** GPOPS-II, CasADi, ACADO Toolkit, Tensorflow*, Pytorch*.
- **Key Courses:** Linear Algebra and Optimization, Optimal Control, Optimization, Probability, Introduction to Machine Learning, Introduction to Reinforcement Learning.

** Elementary proficiency*

Achievements

- Recipient of best paper award at 2025 RAAI, *NTU Singapore.* *2025*
- Recipient of the Prime Minister's Research Fellowship, *Government of India.* *2021 - 2024*
- Secured 1st rank in M.Tech degree specialization, *Birla Institute of Technology Mesra.* *2020*
- Recipient of UG Merit Scholarship, *North-East Council, Government of India.* *2013 - 2017*
- Secured 3rd rank in Indain School Certificate Examination, *Tripura State.* *2013*

Extra-Curriculars

- **Industrial Training:** Led skill development programmes on drone technology for personnel from the Indian Army and Bharat Electronics Limited (BEL).
- **Roles:** BIS Student coordinator (IIT Guwahati), Placement Coordinator & Sponsorship Head (NIT Agartala), Administrator (AgtCovidHelp app).